CO_2 methanation over Ni catalysts supported on CeO_2 and $CeSmO_x$: the influence of Sm addition

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Introduction

Hydrogenation of CO_2 is an attractive process, which can be operated at mild conditions to obtain CH_4 as a valuable synthetic fuel [1]. Compared to noble metals, Ni as a catalyst seems more appropriate for industrial applications due to its lower cost and high activity [2]. Moreover, using CeO_2 as catalytic support appears promising since it can provide strong metal-support interactions with high dispersion of active centers, and its high basicity can promote the adsorption and activation of CO_2 [2]. Adding Sm into CeO_2 can generate oxygen vacancies, promoting its activity in the methanation of CO_2 [2]. Therefore, Ni-based catalysts supported on pure (CeO_2) and Sm-added cerium oxide ($CeSmO_x$) for the target reaction are studied in this work.

Materials and methods

Ni/$CeSmO_x$ was prepared through two methods: a) coprecipitation of $CeSmO_x$ with different Ce/Sm atomic proportions (3:1; 2:1; 1:1) using $(NH_4)_2Ce(NO_3)_6$, $Sm(NO_3)_3{\cdot}6H_2O$ and urea as a precipitant, and a Ni loading of 20 wt.% introduced by the incipient wetness impregnation method using $Ni(NO_3)_2{\cdot}6H_2O$; b) total coprecipitation of Ni/$CeSmO_x$, where $Ni(NO_3)_2{\cdot}6H_2O$, $Ce(NO_3)_3{\cdot}6H_2O$ and $Sm(NO_3)_3{\cdot}6H_2O$ were coprecipitated together using NaOH as a precipitant with the same proportions as in the first method. Accordingly, Ni/CeO_2 was prepared as a reference sample using both methods. The as-prepared catalysts are denoted as Ni/$CeSmO_x$ "*Imp*" and "*Cop*" respectively. The reaction was performed in a continuous fixed-bed tubular reactor employing 0.5 g catalyst at atmospheric pressure with a total flow of 50 ml/min (H_2/CO_2 molar = 4) in 250 ºC – 450 ºC. The products were analyzed using NDIR and FTIR spectrometries. The catalysts were analyzed by TXRF, BET, XRD, HRTEM, XPS, and Raman spectroscopy before and after the reaction.

Results and discussion

The catalysts prepared by total coprecipitation method (*Cop*) show better activity than those prepared by the coprecipitation-impregnation method (*Imp*) at low reaction temperatures (Fig. 1) with higher stability. This could be due to their higher surface areas (*Cop*: 188 – 199 m^2/g; *Imp*: 63 – 114 m^2/g) and smaller nanoparticles (NPs) sizes (Fig. 2). Moreover, *Cop* samples exhibit a higher relative concentration of oxygen vacancies (determined from the Ce^{3+}/Ce^{4+} ratio via XPS), i.e., about 4/6 (*Cop*) compared to 3/7 (*Imp*), promoting their better activity [2]. Overall, CO_2 conversion over Ni/3Ce1SmO$_x$ Cop and Ni/2Ce1SmO$_x$ Cop reaches almost equilibrium at 300 °C, with ca. 89% (Fig. 1), and slightly lower values for the other *Cop* samples, while the *Imp* samples show lower conversion rates mainly owing to their bigger NPs. This trend is strongly emphasized at the lowest reaction temperature (250 °C) where CO_2 conversion drops to 78% and 75% for Ni/2Ce1SmO$_x$ Cop and Ni/3Ce1SmO$_x$ Cop, respectively, and 68% for Ni/CeO_2 Cop and values below 30% for the *Imp* samples. Although Ni/1Ce1SmO$_x$ Cop shows larger amounts of oxygen vacancies due to higher Sm proportions, it is not reflected in the conversion rate of 62 % at 250 °C, which could be related to weaker metal-support interactions owing to its larger Ni NPs (ca. 8.8 nm) and lower oxidized Ni proportion (Ni/1Ce1SmO$_x$ Cop: Ni/NiO: 2/8; Ni/2Ce1SmO$_x$ Cop: Ni/NiO: 1/9) from XPS. In conclusion, Ni/2Ce1SmO$_x$ Cop shows the best performance, probably related to its strong metal-support interaction with small Ni NPs.

References

[1] S. Musab Ahmed, ChemSusChem (2024) e202400310.

[2] G.I. Siakavelas, Applied Catalysis B: Environmental 282, (2021) 119562.

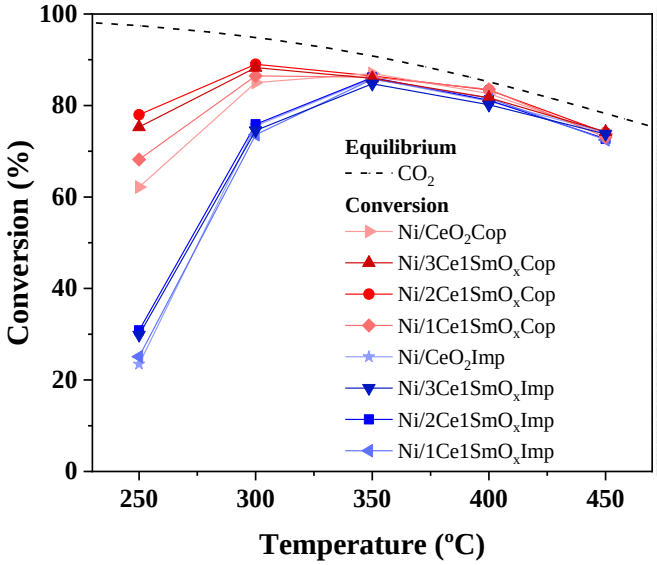

Figure 1. Performance of catalysts

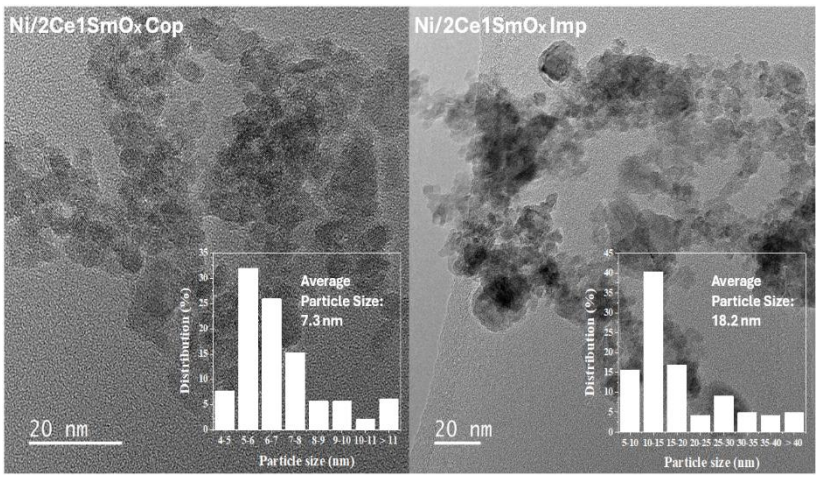

Figure 2. HRTEM images